

C24344923 – Web Development CA

1. Design Document

1.1 Overview

The goal of this project was to create a fully working Library System that allows users to:

- Register an account
- Log in
- Search for books using filters
- Reserve available books
- Manage reserved books

The application demonstrates practical use of:

- PHP programming
- MySQL database
- Session management for security
- CSS styling

I chose an Ancient Library of Alexandria theme to give the library a bit of a personality

1.2 Architecture

The library uses a PHP code structure:

- Shared (header.php and footer.php) to maintain consistent navigation and layout like required.
- General pages (login.php, register.php, reserve.php) to manage sessions, users, and database transactions.
- User interfaces (home.php, index.php, my_reservations.php) displays content retrieved from the database like Books.
- A configuration file (db_config.php) that handles the database connectivity.

1.3 Database Design

The database consists of four primary tables:

Users

- Stores registered user details:
- username (Primary Key)
- password_hash (hashed password)
- email
- mobile (10-digit number validation)
- created_at

Categories

- category_code (Primary Key)
- category_description

Books

- isbn (Primary Key) (Holds all book data)
- title
- author
- publisher
- year_published
- category_code (Foreign key to Categories)
- Reserved_Books

Reserved Books:

- id (Primary Key)
- username (Foreign key → Users)
- isbn (Foreign key → Books)
- reserved_date

Each ISBN is limited to one reservation at a time to prevent double reserving.

2. PHP Pages Used & Descriptions

This section lists every PHP page made, including its use and how it links with other pages.

2.1 home.php

Job: Acts as the main page of the website. Introduces the Library and provides buttons for navigation.

Use:

- Welcomes users.
- Displays buttons linking to the Search page, Login, and Register.
- Accessible to all users regardless of login state.

Interactions:

- Links to index.php (search page).
- Links to login.php and register.php.
- Uses header.php and footer.php for layout.

2.2 index.php (Search Page)

Job: The main working page of the system where users can search and reserve books.

Use:

- Title search
- Author search
- Category filtering using dropdown menu.
- Hides reserved books option.
- Pagination limiting results to 5 books per page.
- Displays availability for books (Available / Reserved).
- Allows logged-in users to reserve available books.

Interactions:

- Uses reserve.php to process reservations.
- Redirects users to login.php if they attempt to reserve while not logged in.
- Uses db_config.php to fetch books and categories.
- Includes header.php and footer.php.
- Uses session variables for login-based actions.
- Links to my_reservations.php via the navigation menu.

2.3 header.php

Job: Provides a shared site header with navigation links.

Use:

- Displays library logo and title.
- Shows menu items (Home, Search, Login/Register (Not Logged In) OR My Reservations/Logout (Logged In)).
- Displays the logged-in username.

Interactions:

- Included on all pages.
- Accesses PHP session to determine navigation state.

2.4 footer.php

Job: Provides a consistent footer section.

Use:

- Shows copyright.
- Matches header styling.
- Closes page layout elements.

Interactions:

- Included on all pages.

2.5 login.php

Job: Handles user login.

Use:

- Accepts username and password.
- Validates using the database using `password_verify()`.
- Sets a session start on successful login.
- Redirects users to `index.php` after login.
- Displays error messages if log in fails.

Interactions:

- Uses `db_config.php` to access users database.
- Redirects to `index.php` on success.
- Uses `header.php` and `footer.php`.

2.6 register.php

Job: Creates new user accounts.

Use:

- Enforces validation:
 1. 10-digit mobile number
 2. Password minimum length of 6
 3. Matching password confirmation
 4. Unique usernames
 5. Stores passwords using `password_hash()`.
 6. Redirects to `login.php` upon successful registration.

Interactions:

- Send data to Users table using `db_config.php`.
- Redirects to `login.php`.

- Uses header.php and footer.php.

2.7 logout.php

Job: Ends the session.

Use:

- Clears session data.
- Redirects back to the home page.

Interactions:

- Uses PHP sessions.

2.8 reserve.php

Job: Handles reservations.

Use:

- Checks that the user is logged in.
- Uses SQL statements to insert reservations.
- Prevents double reservations due to unique ISBN limits.
- Stores reservation time.
- Redirects user back to the search page (index.php).

Interactions:

- Uses db_config.php.
- Called only from index.php.

2.9 my_reservations.php

Job: Allows users to view and manage their reservations.

Use:

- Displays all reservations by the logged in user.
- Allows removal of any reservation.
- Refreshes data automatically after removal.

Interactions:

- Reads from reserved_books and books tables using db_config.php.
- Uses header.php and footer.php.

2.10 db_config.php

Job: Maintains MySQL connection credentials.

Use:

- Sets up database connection using MySQL.
- Used by all PHP files needing database access.

3. Validation

All input validation is performed server side:

Registration needs completion of all fields.

- Mobile numbers needs to be numeric and 10 digits long.
- Password has be at least 6 characters.
- Double password confirmation must be the same.

Username must be unique.

- Password validation uses PHP's password_hash() and password_verify().

4. Security

Security is implemented using these methods:

- Prepared SQL statements prevent SQL injection.
- Password hashing makes sure plain-text passwords are never stored.
- Session protection ensures restricted pages cannot be accessed without login.
- Reservation and deletion actions are user restricted (only users can modify their own reservations.)

5. Search & Reservation

Users can search by:

- Book title
- Author name
- Category dropdown

Search results:

- Display a maximum of 5 books per page.
- Show whether books are Available or Reserved.
- Allow bookings only if a book is available.
- Hide option removes reserved books from display.

Users manage active reservations through My Reservations, where books may be removed.

6. UI Design

The interface adds an Ancient Alexandria theme which includes:

- Soft light colors
- Scroll-styled icons/emojis
- Inline-aligned tables and buttons

All pages share:

- A common navigation header (Different when logged in and not)
- A common footer.
- A consistent page layout for usability.

7. Testing

Testing verified:

- Registration works correctly.
- Login works with correct information.
- Incorrect login information generate error message.
- Book searching and filtering works across all field combinations.
- Pagination limits results to 5 per page.
- Reservation prevents the reservation of already reserved books.
- Reservation removal in My Reservations.
- Navigation from all screens.

8. Conclusion

This Library System completes all project requirements specified in the brief. It shows secure PHP programming principles, database integration, complete server-side validation, and functional UI .

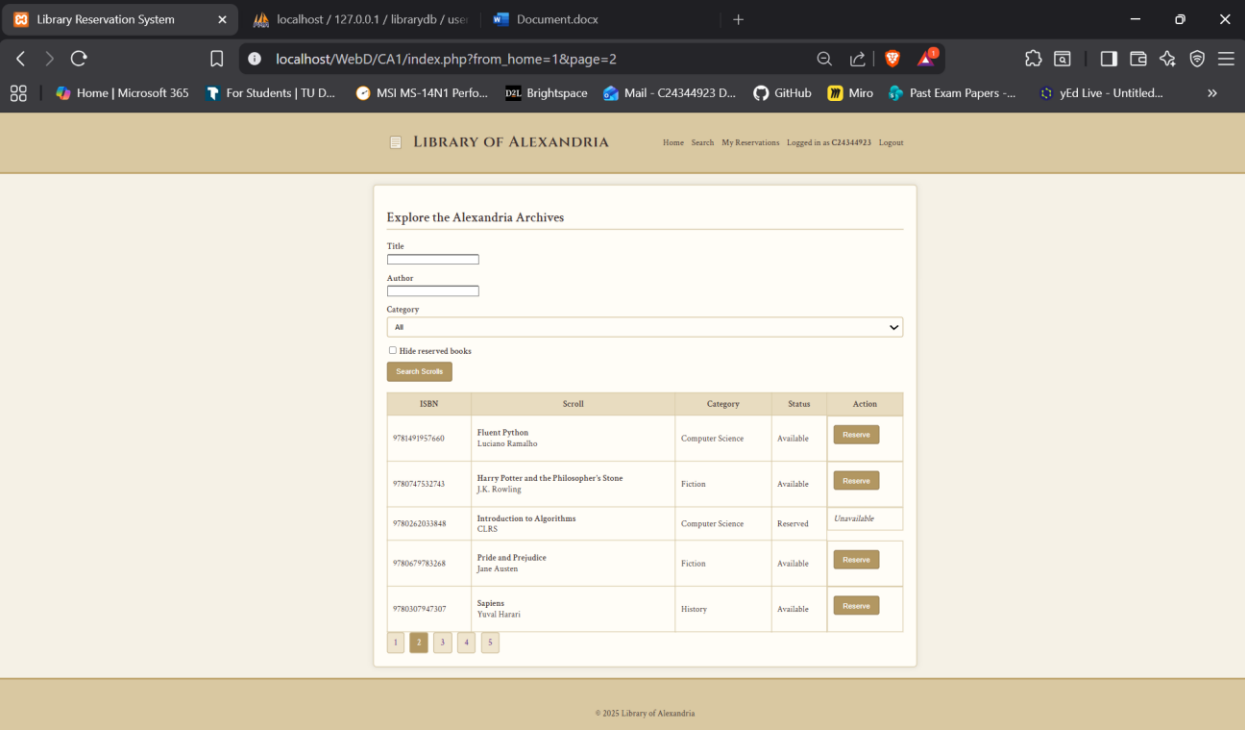
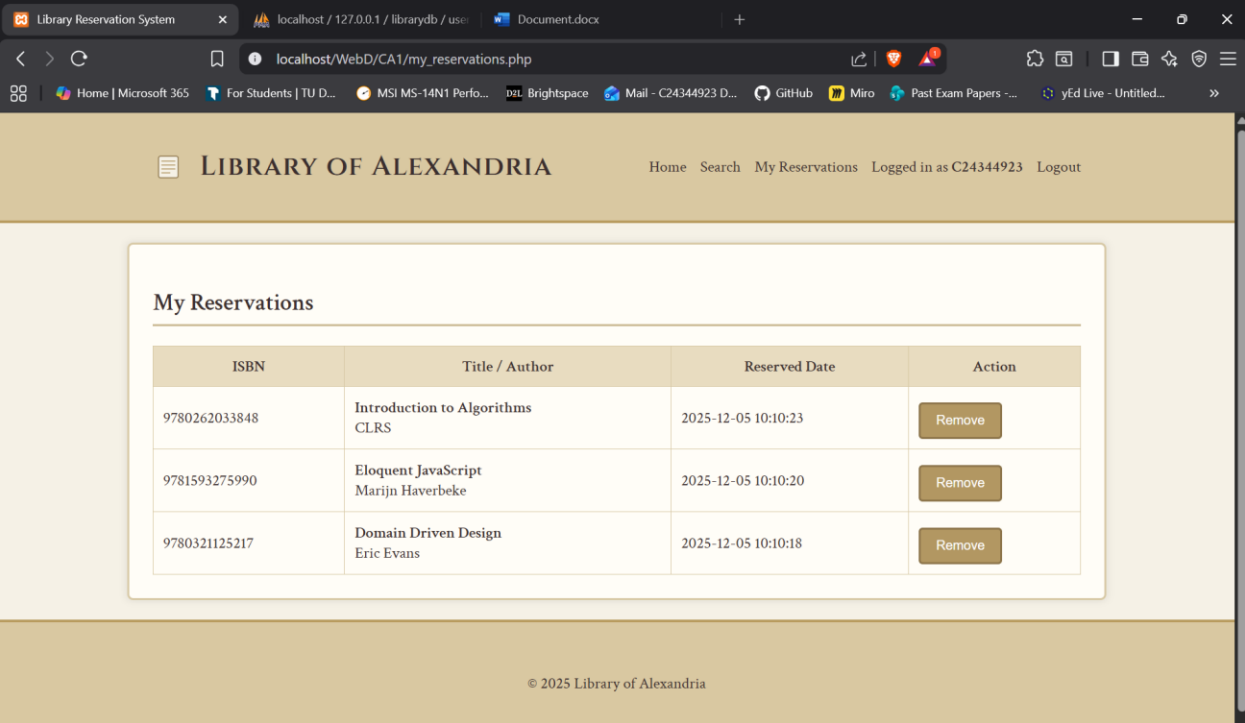
9. Website Pages (Screenshots)

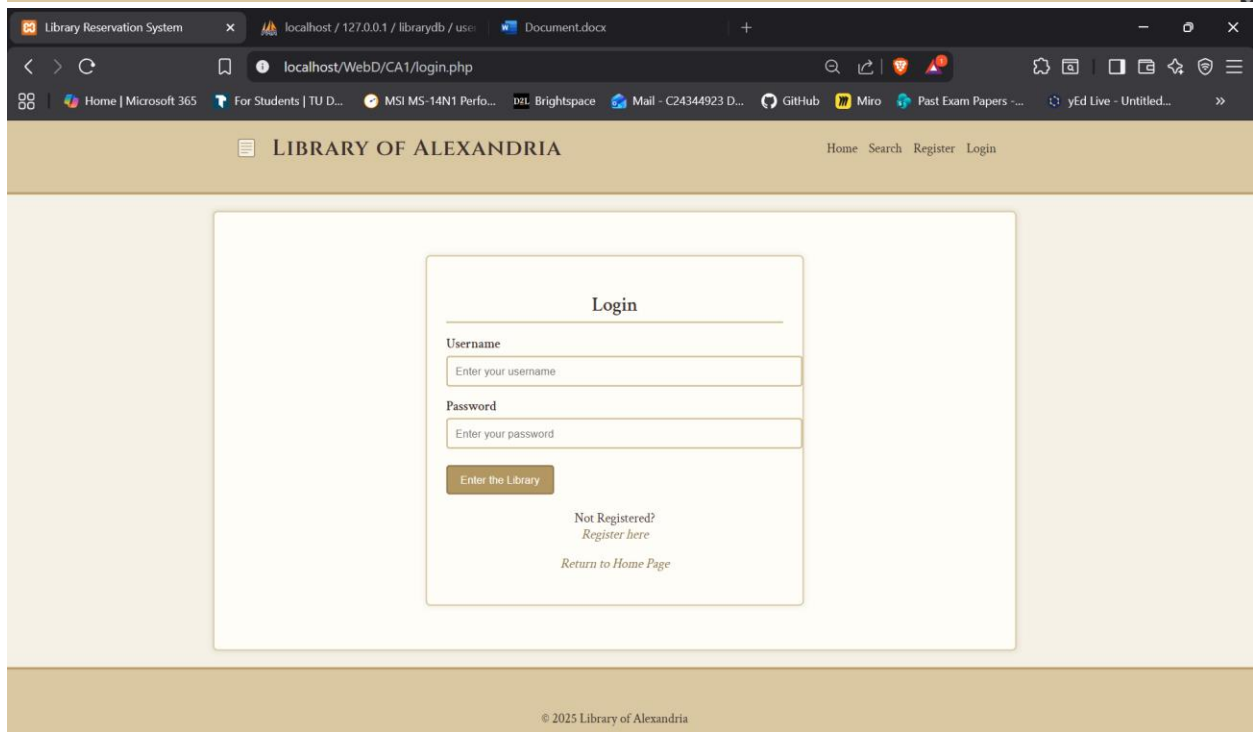
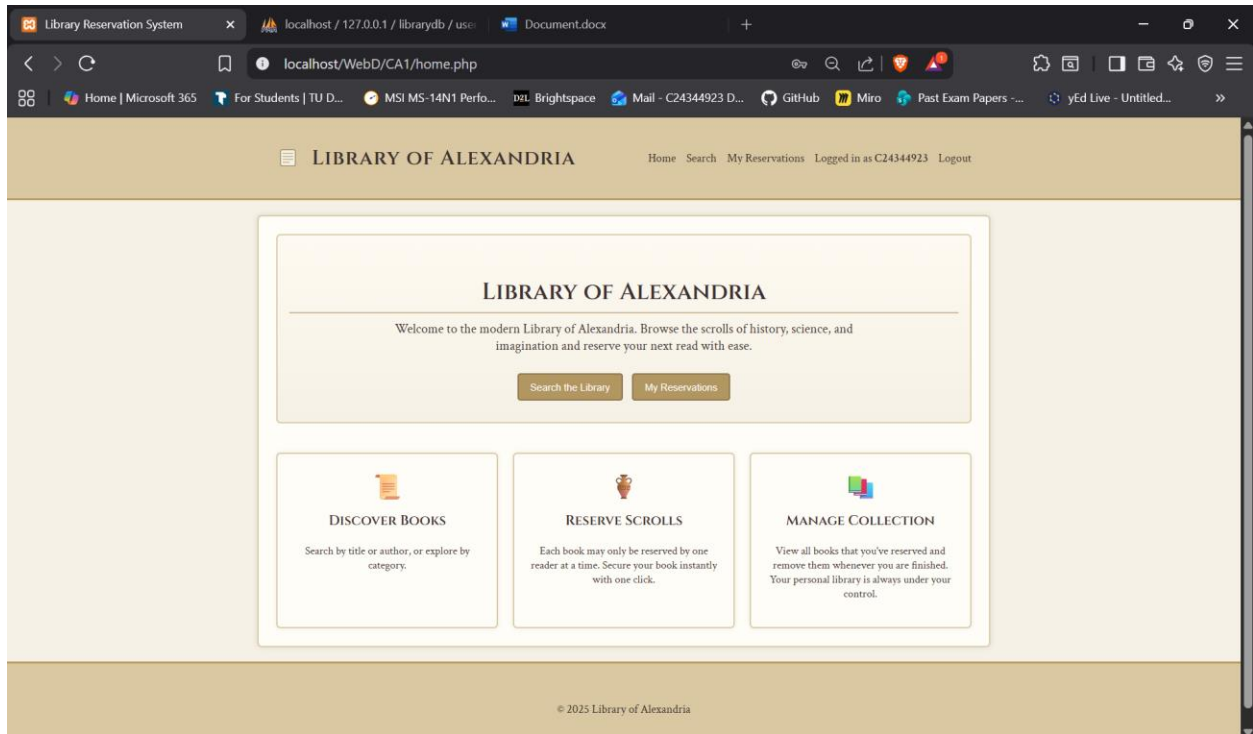
I couldn't sort the screenshots so this is the order:

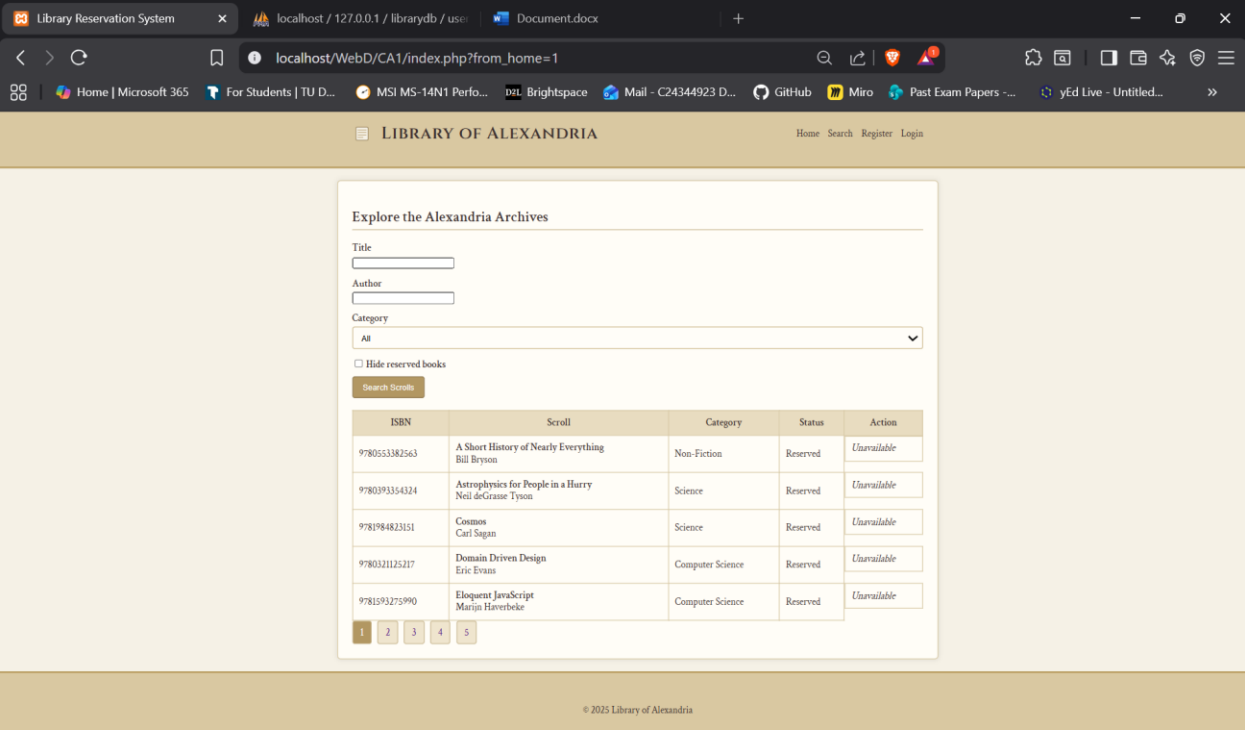
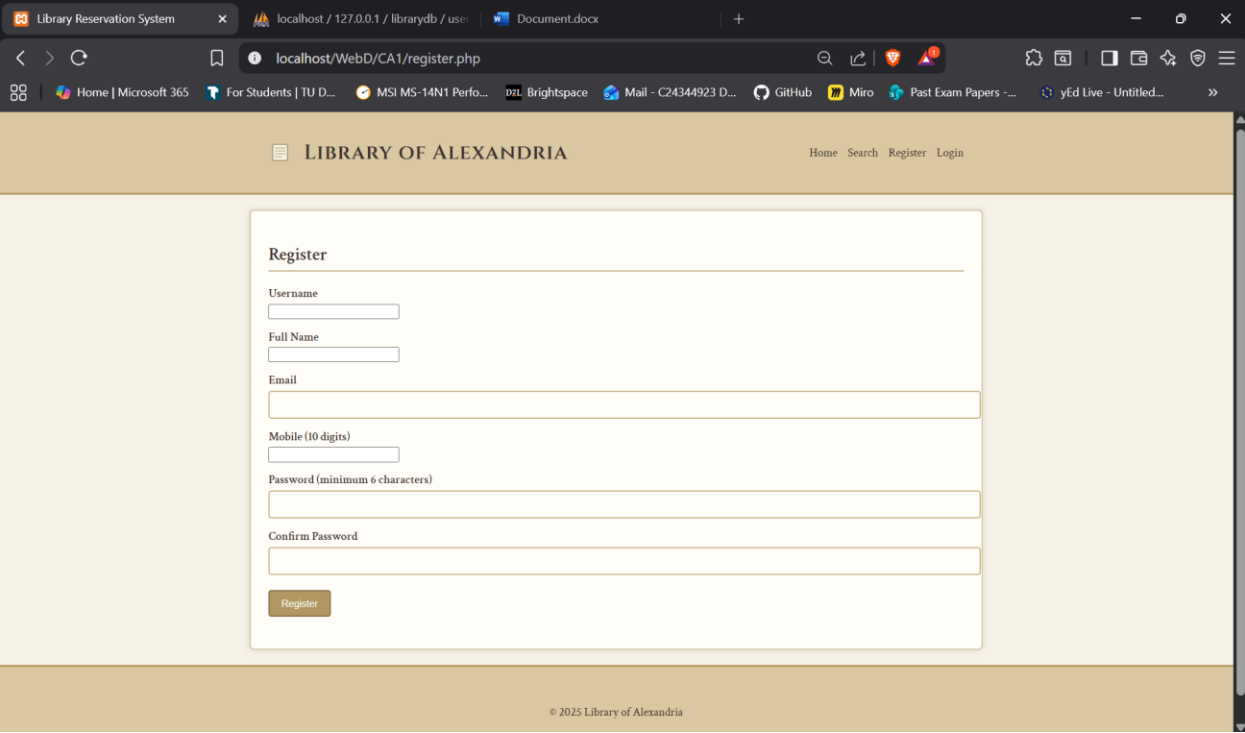
1. my_reservation.php (Reservations)
2. Index.php (Search – Logged In)
3. Home.php (Logged In)
4. Login.php (Log In page)
5. Register.php (Register page)
6. Index.php (Search – Logged Out)
7. Home.php (Logged Out)

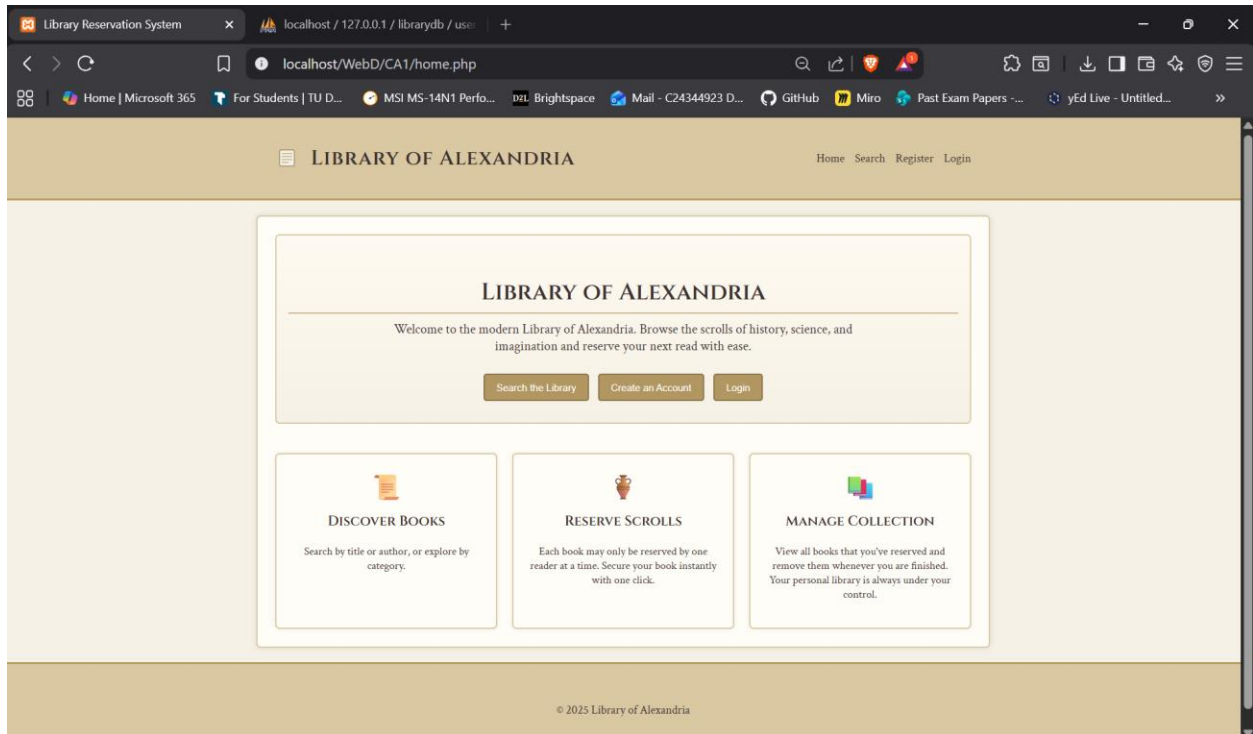
The file names are in the search bar.

The screenshots show a comparison between logged in and logged out.









10. Code (Link)

[Code Folder](#)

11. Code (Screenshots – Just Incase)

```

34     $ins->close();
35 } catch (Exception $e) {
36     $_SESSION['flash_error'] = "An error occurred while reserving the book.";
37 }
38
39 $mysqli->close();
40 header("Location: " . $return_url);
41 exit;
42 |

```

hp library.sql login.php logout.php my_reservations.php register.php reserve.php X ▾ ▾

C: > xampp > htdocs > WebD > CA1 > reserve.php

```

1  <?php
2  require_once 'db_config.php';
3  session_start();
4
5  if (empty($_SESSION['username'])) {
6      header("Location: login.php");
7      exit;
8  }
9
10 if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
11     header("Location: index.php");
12     exit;
13 }
14
15 $isbn = $_POST['isbn'] ?? '';
16 $return_url = $_POST['return_url'] ?? 'index.php';
17
18 if ($isbn === '') {
19     $_SESSION['flash_error'] = "No book selected.";
20     header("Location: " . $return_url);
21     exit;
22 }
23 |
24 $mysqli = db_connect();
25
26 try {
27     $ins = $mysqli->prepare("INSERT INTO reserved_books (username, isbn, reserved_date) VALUES (?, ?, NOW(
28     $ins->bind_param('ss', $_SESSION['username'], $isbn);
29     if ($ins->execute()) {
30         $_SESSION['flash_success'] = "Book reserved successfully.";
31     } else {
32         $_SESSION['flash_error'] = "Could not reserve book. It may already be reserved.";
33     }
34 }
35
36
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100
101
102
103
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105
106
107
108
109
110
111
112
113
114
115
116
117 </div>
118
119 <input type="submit" value="Register">
120
121 </form>
122
123 <?php include 'footer.php'; ?>
124

```

```

96         <label>Username</label>
97         <input name="username"
98             value="<?php echo htmlspecialchars($_POST['username'] ?? '') ?>">
99     </div>
100
101     <div class="form-group">
102         <label>Full Name</label>
103         <input name="fullname"
104             value="<?php echo htmlspecialchars($_POST['fullname'] ?? '') ?>">
105     </div>
106
107     <div class="form-group">
108         <label>Email</label>
109         <input name="email" type="email"
110             value="<?php echo htmlspecialchars($_POST['email'] ?? '') ?>">
111     </div>
112
113     <div class="form-group">
114         <label>Mobile (10 digits)</label>
115         <input name="mobile"
116             value="<?php echo htmlspecialchars($_POST['mobile'] ?? '') ?>">
117     </div>
118
119     <div class="form-group">
120         <label>Password (minimum 6 characters)</label>
121         <input name="password" type="password">
122     </div>
123
124     <div class="form-group">
125         <label>Confirm Password</label>
126         <input name="password_confirm" type="password">

```

```

66
67         if ($insert->execute()) {
68             // ✅ SUCCESS -> GO TO LOGIN PAGE
69             $insert->close();
70             $mysqli->close();
71             header("Location: login.php");
72             exit;
73         } else {
74             $errors[] = "Registration failed. Please try again.";
75             $insert->close();
76         }
77     }
78     $mysqli->close();
79 }
80 }
81
82 include 'header.php';
83 ?>
84
85 <h2>Register</h2>
86
87 <?php if ($errors): ?>
88     <div class="alert">
89         <?php echo implode("<br>", array_map("htmlspecialchars", $errors)); ?>
90     </div>
91 <?php endif; ?>
92
93 <form method="post" action="register.php" novalidate>
94
95     <div class="form-group">

```

```

34     $errors[] = "Passwords do not match.";
35 }
36
37 if (!$errors) {
38     $mysqli = db_connect();
39
40     /* Check username uniqueness */
41     $stmt = $mysqli->prepare("SELECT username FROM users WHERE username = ?");
42     $stmt->bind_param('s', $username);
43     $stmt->execute();
44     $stmt->store_result();
45
46     if ($stmt->num_rows > 0) {
47         $errors[] = "Username is already in use.";
48         $stmt->close();
49     } else {
50         $stmt->close();
51
52         $hash = password_hash($password, PASSWORD_DEFAULT);
53         $insert = $mysqli->prepare("
54             INSERT INTO users
55             (username, fullname, email, mobile, password_hash)
56             VALUES (?, ?, ?, ?, ?)
57         ");
58         $insert->bind_param(
59             'sssss',
60             $username,
61             $fullname,
62             $email,
63             $mobile,
64             $hash
65         );

```

C: > xampp > htdocs > WebD > CA1 > 🐞 register.php

```
1  <?php
2  require_once 'db_config.php';
3  session_start();
4
5  $errors = [];
6
7  if ($_SERVER['REQUEST_METHOD'] === 'POST') {
8      $username = trim($_POST['username'] ?? '');
9      $fullname = trim($_POST['fullname'] ?? '');
10     $email = trim($_POST['email'] ?? '');
11     $mobile = trim($_POST['mobile'] ?? '');
12     $password = $_POST['password'] ?? '';
13     $confirm = $_POST['password_confirm'] ?? '';
14
15     /* Validation */
16     if ($username === '' || $fullname === '' || $email === '' ||
17         $mobile === '' || $password === '' || $confirm === '') {
18         $errors[] = "All fields are required.";
19     }
20
21     if (!preg_match('/^[0-9]{10}$/', $mobile)) {
22         $errors[] = "Mobile number must be numeric and exactly 10 digits.";
23     }
24
25     if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
26         $errors[] = "Invalid email format.";
27     }
28
29     if (strlen($password) < 6) {
30         $errors[] = "Password must be at least 6 characters long.";
31     }
32
33     if ($password !== $confirm) {
```

C: > xampp > htdocs > WebD > CA1 > 🐞 my_reservations.php

```
99
100 <?php
101 $mysqli->close();
102 include 'footer.php';
103 ?>
104
```

```

66
67 <?php if (empty($reservations)): ?>
68     <p>You have no current reservations.</p>
69 <?php else: ?>
70     <table class="table">
71         <thead>
72         <tr>
73             <th>ISBN</th>
74             <th>Title / Author</th>
75             <th>Reserved Date</th>
76             <th>Action</th>
77         </tr>
78         </thead>
79         <tbody>
80         <?php foreach ($reservations as $r): ?>
81             <tr>
82                 <td><?php echo htmlspecialchars($r['isbn']); ?></td>
83                 <td>
84                     <strong><?php echo htmlspecialchars($r['title']); ?></strong><br>
85                     <span><?php echo htmlspecialchars($r['author']); ?></span>
86                 </td>
87                 <td><?php echo htmlspecialchars($r['date']); ?></td>
88                 <td>
89                     <form method="post" style="display:inline;" onsubmit="return confirm('Remove this rese
90                         <input type="hidden" name="remove_id" value="<?php echo (int)$r['id']; ?>">
91                         <input type="submit" value="Remove">
92                     </form>
93                 </td>
94             </tr>
95         <?php endforeach; ?>
96     </tbody>
97 </table>
98 <?php endif; ?>

```

```

34     FROM reserved_books rb
35     JOIN books b ON rb.isbn = b.isbn
36     WHERE rb.username = ?
37     ORDER BY rb.reserved_date DESC
38 "];
39 $stmt->bind_param('s', $username);
40 $stmt->execute();
41 $stmt->bind_result($id, $isbn, $title, $author, $date);
42
43 $reservations = [];
44 while ($stmt->fetch()) {
45     $reservations[] = [
46         'id' => $id,
47         'isbn' => $isbn,
48         'title' => $title,
49         'author' => $author,
50         'date' => $date
51     ];
52 }
53 $stmt->close();
54
55 include 'header.php';
56 ?>
57
58 <h2>My Reservations</h2>
59
60 <?php if ($flash_error): ?>
61     <div class="alert"><?php echo htmlspecialchars($flash_error); ?></div>
62 <?php endif; ?>
63 <?php if ($flash_success): ?>
64     <div class="success"><?php echo htmlspecialchars($flash_success); ?></div>
65 <?php endif; ?>

```

```
er.php  home.php X index.php library.sql login.php logout.php my_reservations.php X
C: > xampp > htdocs > WebD > CA1 > my_reservations.php
1  <?php
2  require_once 'db_config.php';
3  session_start();
4
5  if (empty($_SESSION['username'])) {
6      header("Location: login.php");
7      exit;
8  }
9
10 $mysqli = db_connect();
11 $username = $_SESSION['username'];
12
13 // Flash from remove
14 $flash_error = $_SESSION['flash_error'] ?? '';
15 $flash_success = $_SESSION['flash_success'] ?? '';
16 unset($_SESSION['flash_error'], $_SESSION['flash_success']);
17
18 // Handle removal
19 if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['remove_id'])) {
20     $id = (int)$_POST['remove_id'];
21     $del = $mysqli->prepare("DELETE FROM reserved_books WHERE id = ? AND username = ?");
22     $del->bind_param('is', $id, $username);
23     if ($del->execute()) {
24         $flash_success = "Reservation removed.";
25     } else {
26         $flash_error = "Could not remove reservation.";
27     }
28     $del->close();
29 }
30
31 // Fetch reservations
32 $stmt = $mysqli->prepare("
33     SELECT rb.id, rb.isbn, b.title, b.author, rb.reserved_date
34     FROM reserved_books rb
35     JOIN books b ON rb.isbn = b.isbn
36     ORDER BY rb.reserved_date DESC
37 ");
38 $stmt->execute();
39 $result = $stmt->get_result();
40 $reservations = [];
41 while ($row = $result->fetch_assoc()) {
42     $reservations[] = $row;
43 }
44 $stmt->close();
45
```

```
footer.php header.php home.php index.php library.sql login.php logout.php X
C: > xampp > htdocs > WebD > CA1 > logout.php
1  <?php
2  session_start();
3  session_destroy();
4  header("Location: home.php");
5  exit;
6  ?>
7
```

```

126         <a href="register.php">Register here</a>
127     </p>
128
129     <p>
130         <a href="home.php">Return to Home Page</a>
131     </p>
132
133 </div>
134
135 </div>
136
137 <?php include 'footer.php'; ?>
138

```

```

96     </div>
97 <?php endif; ?>
98
99 <form method="post" action="login.php">
100
101     <div class="form-group">
102         <label>Username</label>
103         <input name="username"
104             type="text"
105             placeholder="Enter your username"
106             value="<?php echo htmlspecialchars($_POST['username']) ?? ''; ?>"
107             required>
108     </div>
109
110     <div class="form-group">
111         <label>Password</label>
112         <input name="password"
113             type="password"
114             placeholder="Enter your password"
115             required>
116     </div>
117
118     <input type="submit" value="Enter the Library">
119
120 </form>
121
122 <div class="login-links">
123
124     <p>
125         Not Registered? <br>

```

```
64     margin-bottom: 16px;
65     color: #555;
66 }
67
68 .login-links {
69     margin-top: 12px;
70     text-align: center;
71 }
72
73 .login-links a {
74     color: #8c744a;
75     text-decoration: none;
76     font-style: italic;
77 }
78
79 .login-links a:hover {
80     text-decoration: underline;
81 }
82
83 </style>
84
85 <div class="login-card">
86
87     <h2>Login</h2>
88
89     <p class="login-subtitle">
90
91     </p>
92
93     <?php if ($errors): ?>
94         <div class="alert">
95             <?php echo implode('<br>', array_map('htmlspecialchars', $errors)); ?>
```

```
33     }
34
35     $stmt->close();
36     $mysqli->close();
37 }
38 }
39
40 include 'header.php';
41 ?>
42
43 <style>
44 /* Login page custom layout */
45
46 .login-card {
47     max-width: 420px;
48     margin: 30px auto;
49     background: #ffffd7;
50     border: 2px solid #d8c9a5;
51     border-radius: 6px;
52     padding: 24px;
53     box-shadow: 0 0 8px rgba(0,0,0,0.08);
54 }
55
56 .login-card h2 {
57     text-align: center;
58     margin-bottom: 8px;
59 }
60
61 .login-subtitle {
62     text-align: center;
63     font-size: 14px;
```

db_config.php footer.php header.php home.php index.php library.sql login.php X

C: > xampp > htdocs > WebD > CA1 > login.php

```
1  <?php
2  require_once 'db_config.php';
3  session_start();
4
5  $errors = [];
6
7  if ($_SERVER['REQUEST_METHOD'] === 'POST') {
8      $username = trim($_POST['username'] ?? '');
9      $password = $_POST['password'] ?? '';
10
11      if ($username === '' || $password === '') {
12          $errors[] = "Both username and password are required.";
13      } else {
14          $mysqli = db_connect();
15          $stmt = $mysqli->prepare("SELECT password_hash FROM users WHERE username = ?");
16          $stmt->bind_param('s', $username);
17          $stmt->execute();
18          $stmt->store_result();
19
20          if ($stmt->num_rows === 1) {
21              $stmt->bind_result($hash);
22              $stmt->fetch();
23
24              if (password_verify($password, $hash)) {
25                  $_SESSION['username'] = $username;
26                  header("Location: index.php");
27                  exit;
28              } else {
29                  $errors[] = "Invalid username or password.";
30              }
31          } else {
32              $errors[] = "Invalid username or password.";
33          }
34      }
35  }
```

db_config.php footer.php header.php home.php index.php library.sql X

C: > xampp >htdocs > WebD > CA1 > library.sql

```
64 ('9780192804587','The History of the Ancient World','Susan Bauer',5,'Norton',2007),
65 ('9780307947307','Sapiens','Yuval Harari',5,'Vintage',2015),
66
67 ('9780393354324','Astrophysics for People in a Hurry','Neil deGrasse Tyson',6,'WW Norton',2017),
68 ('9780465030798','The Elegant Universe','Brian Greene',6,'Vintage',2003),
69 ('9780307387842','The Gene','Siddhartha Mukherjee',6,'Scribner',2016),
70 ('9781984823151','Cosmos','Carl Sagan',6,'Random House',1980),
71 ('9780345803481','The Immortal Life of Henrietta Lacks','Rebecca Skloot',6,'Broadway',2010);
72
```

```
33 UNIQUE (isbn),
34 FOREIGN KEY (username) REFERENCES users(username),
35 FOREIGN KEY (isbn) REFERENCES books(isbn)
36 );
37
38 INSERT INTO categories (category_description) VALUES
39 ('Fiction'),('Non-Fiction'),('Business'),
40 ('Computer Science'),('History'),('Science');
41
42 /* FULL 25 BOOKS */
43 INSERT INTO books (isbn, title, author, category_code, publisher, year_published) VALUES
44 ('9780140449136','The Odyssey','Homer',1,'Penguin',1996),
45 ('9780679783268','Pride and Prejudice','Jane Austen',1,'Vintage',2000),
46 ('9780743273565','The Great Gatsby','Fitzgerald',1,'Scribner',2004),
47 ('9780385348119','The Martian','Andy Weir',1,'Crown',2014),
48 ('9780099406133','The Lord of the Rings','Tolkien',1,'HarperCollins',1954),
49 ('9780747532743','Harry Potter and the Philosopher''s Stone','J.K. Rowling',1,'Bloomsbury',1997),
50 ('9780062316110','The Alchemist','Paulo Coelho',1,'HarperOne',1988),
51 ('9780307277671','The Road','Cormac McCarthy',1,'Vintage',2006),
52
53 ('9780374533557','Thinking, Fast and Slow','Daniel Kahneman',2,'FSG',2011),
54 ('9780553382563','A Short History of Nearly Everything','Bill Bryson',2,'Broadway Books',2003),
55 ('9780307465351','The Lean Startup','Eric Ries',3,'Crown',2011),
56 ('9781451648539','Steve Jobs','Walter Isaacson',3,'Simon & Schuster',2011),
57 ('9781593275990','Eloquent JavaScript','Marijn Haverbeke',4,'No Starch Press',2014),
58 ('9780131103627','The C Programming Language','K&R',4,'Prentice Hall',1988),
59 ('9780321125217','Domain Driven Design','Eric Evans',4,'Addison-Wesley',2003),
60 ('9780262033848','Introduction to Algorithms','CLRS',4,'MIT Press',2009),
61 ('9781491957660','Fluent Python','Luciano Ramalho',4,'O'Reilly',2015),
62
63 ('9780140104765','The Art of War','Sun Tzu',5,'Penguin',2003),
64 ('9780192804587','The History of the Ancient World','Susan Bauer',5,'Norton',2007),
```

C: > xampp > htdocs > WebD > CA1 > library.sql

```
1  CREATE DATABASE IF NOT EXISTS librarydb CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
2  USE librarydb;
3
4  CREATE TABLE users (
5      username VARCHAR(50) PRIMARY KEY,
6      fullname VARCHAR(100),
7      email VARCHAR(100),
8      mobile VARCHAR(10),
9      password_hash VARCHAR(255),
10     created_at DATETIME DEFAULT CURRENT_TIMESTAMP
11 );
12
13 CREATE TABLE categories (
14     category_code INT PRIMARY KEY AUTO_INCREMENT,
15     category_description VARCHAR(100)
16 );
17
18 CREATE TABLE books (
19     isbn VARCHAR(20) PRIMARY KEY,
20     title VARCHAR(255),
21     author VARCHAR(200),
22     category_code INT,
23     publisher VARCHAR(200),
24     year_published YEAR,
25     FOREIGN KEY (category_code) REFERENCES categories(category_code)
26 );
27
28 CREATE TABLE reserved_books (
29     id INT PRIMARY KEY AUTO_INCREMENT,
30     username VARCHAR(50),
31     isbn VARCHAR(20),
32     reserved_date DATETIME DEFAULT CURRENT_TIMESTAMP,
33     UNIQUE (isbn),
```

```

190 <?php else: ?>
191 <em>Unavailable</em>
192 <?php endif; ?>
193 </td>
194 </tr>
195
196 <?php endwhile; ?>
197 </table>
198
199 <!-- PAGINATION -->
200 <div class="pagination">
201 <?php for($p=1;$p<=$pages;$p++): ?>
202 <a href="?from_home=1&page=<?=$p?>" class="<?=$p==$page?'current':''?>">
203 <?=$p?>
204 </a>
205 <?php endfor; ?>
206 </div>
207
208 <?php
209 $stmt->close();
210 $mysqli->close();
211 include 'footer.php';
212 ?>
213

```

```

158 <?php while($row = $result->fetch_assoc()): ?>
159 <?php
160 $isbn = $row['isbn'];
161 $r = $mysqli->prepare("SELECT COUNT(*) FROM reserved_books WHERE isbn=?");
162 $r->bind_param("s",$isbn);
163 $r->execute();
164 $r->bind_result($resCount);
165 $r->fetch();
166 $r->close();
167 $busy = $resCount > 0;
168 ?>
169
170 <tr>
171 <td><?=$isbn?></td>
172 <td><strong><?=$row['title']?></strong><br><?=$row['author']?></td>
173 <td><?=$catMap[$row['category_code']]?></td>
174
175 <td>
176 <span class="status <?=$busy ? 'reserved':'available'?>">
177 <?=$busy?'Reserved':'Available'?>
178 </span>
179 </td>
180
181 <td class="actions">
182 <?php if(!$busy && isset($_SESSION['username'])): ?>
183 <form method="post" action="reserve.php">
184 <input type="hidden" name="isbn" value="<?=$isbn?>">
185 <input type="hidden" name="return_url" value="<?=htmlspecialchars($_SERVER['REQUEST_URI'])?>">
186 <button>Reserve</button>
187 </form>
188 <?php elseif(!$busy): ?>
189 <em>Login to reserve</em>

```

```
126 <div class="form-group">
127   <label>Category</label>
128   <select name="category">
129     <option value="0">All</option>
130     <?php foreach($cats as $c): ?>
131       <option value="<?=$c['category_code']?>"
132         <?php if ($category==$c['category_code']) echo "selected"; ?>>
133         <?=$c['category_description']?>
134       </option>
135     <?php endforeach; ?>
136   </select>
137 </div>
138
139 <label>
140   <input type="checkbox" name="hide_reserved" value="1"
141     <?php if($hideReserved) echo 'checked'; ?>>
142   Hide reserved books
143 </label>
144
145 <input type="submit" value="Search Scrolls">
146
147 </form>
148
149 <table class="table">
150 <tr>
151 <th>ISBN</th>
152 <th>Scroll</th>
153 <th>Category</th>
154 <th>Status</th>
155 <th>Action</th>
156 </tr>
157
```

```

96 include 'header.php';
97 ?>
98
99 <!-- ----- LIBRARY PAGE UI ----- -->
100
101 <h2>Explore the Alexandria Archives</h2>
102
103 <?php if ($flash_error): ?>
104 |   <div class="alert"><?php echo htmlspecialchars($flash_error); ?></div>
105 <?php endif; ?>
106
107 <?php if ($flash_success): ?>
108 |   <div class="success"><?php echo htmlspecialchars($flash_success); ?></div>
109 <?php endif; ?>
110
111 <form method="get">
112
113 <!-- ✅ KEEP SPECIAL PARAMETER SO HOME REDIRECT IS SKIPPED -->
114 <input type="hidden" name="from_home" value="1">
115
116 <div class="form-group">
117 |   <label>Title</label>
118 |   <input name="title" value="<?htmlspecialchars($title)?>">
119 </div>
120
121 <div class="form-group">
122 |   <label>Author</label>
123 |   <input name="author" value="<?htmlspecialchars($author)?>">
124 </div>
125
126 <div class="form-group">
127 |   <label>Category</label>

```

```

64 $whereSQL = $where ? "WHERE " . implode(" AND ", $where) : "";
65
66 /* Count rows */
67 $count = $mysqli->prepare("SELECT COUNT(*) FROM books $whereSQL");
68 if ($types) $count->bind_param($types, ...$params);
69 $count->execute();
70 $count->bind_result($total);
71 $count->fetch();
72 $count->close();
73
74 $pages = max(1, ceil($total / $limit));
75
76 /* Fetch records */
77 $sql = "SELECT isbn,title,author,category_code
78     FROM books
79     $whereSQL
80     ORDER BY title
81     LIMIT ? OFFSET ?";
82
83 $stmt = $mysqli->prepare($sql);
84 $bindTypes = $types . 'ii';
85 $bindParams = array_merge($params, [$limit,$offset]);
86 $stmt->bind_param($bindTypes, ...$bindParams);
87 $stmt->execute();
88 $result = $stmt->get_result();
89
90 /* Category Map */
91 $catMap = [];
92 foreach ($cats as $c) {
93     $catMap[$c['category_code']] = $c['category_description'];
94 }
95

```

```
31 $title = trim($_GET['title'] ?? '');
32 $author = trim($_GET['author'] ?? '');
33 $category = $_GET['category'] ?? '0';
34 $hideReserved = isset($_GET['hide_reserved']);
35
36 $page = max(1, (int)($_GET['page'] ?? 1));
37 $limit = 5;
38 $offset = ($page - 1) * $limit;
39
40 /* WHERE building */
41 $where = [];
42 $params = [];
43 $types = '';
44
45 if ($title !== '') {
46     $where[] = "title LIKE ?";
47     $params[] = "%$title%";
48     $types .= 's';
49 }
50 if ($author !== '') {
51     $where[] = "author LIKE ?";
52     $params[] = "%$author%";
53     $types .= 's';
54 }
55 if ($category !== '0') {
56     $where[] = "category_code = ?";
57     $params[] = (int)$category;
58     $types .= 'i';
59 }
60 if ($hideReserved) {
61     $where[] = "isbn NOT IN (SELECT isbn FROM reserved_books)";
62 }
63
```

db_config.php

footer.php

header.php

home.php

index.php X



C: > xampp > htdocs > WebD > CA1 > index.php

```
1  <?php
2  // -----
3  // HOME ROUTING LOGIC
4  // If the site is opened directly with no parameters
5  // send user to HOME page.
6  // -----
7  if (!isset($_GET['from_home'])) {
8      header("Location: home.php");
9      exit;
10 }
11
12 // Continue with library/search page below
13 require_once 'db_config.php';
14 session_start();
15
16 $mysqli = db_connect();
17
18 /* Flash messages */
19 $flash_error  = $_SESSION['flash_error'] ?? '';
20 $flash_success = $_SESSION['flash_success'] ?? '';
21 unset($_SESSION['flash_error'], $_SESSION['flash_success']);
22
23 /* Categories */
24 $cats = [];
25 $cq = $mysqli->query("SELECT category_code, category_description FROM categories ORDER BY category_descrip");
26 while ($row = $cq->fetch_assoc()) {
27     $cats[] = $row;
28 }
29
30 /* Search inputs */
```

```

98 <div class="feature-grid">
99
100 <div class="feature">
101 <div class="feature-icon">📖</div>
102 <h3>Discover Books</h3>
103 <p>
104     Search by title or author, or explore by category.
105 </p>
106 </div>
107
108 <div class="feature">
109 <div class="feature-icon">🔒</div>
110 <h3>Reserve Scrolls</h3>
111 <p>
112     Each book may only be reserved by one reader at a time. Secure your
113     book instantly with one click.
114 </p>
115 </div>
116
117 <div class="feature">
118 <div class="feature-icon">🗑️</div>
119 <h3>Manage Collection</h3>
120 <p>
121     View all books that you've reserved and remove them whenever you are finished.
122     Your personal library is always under your control.
123 </p>
124 </div>
125
126 </div>
127
128 <?php include 'footer.php'; ?>
129

```

```
66 </style>
67
68 <div class="hero">
69   <h2>Library of Alexandria</h2>
70
71   <p>
72     Welcome to the modern Library of Alexandria.
73     Browse the scrolls of history, science, and imagination and reserve your next read
74     with ease.
75   </p>
76
77   <div class="cta-buttons">
78     <a href="index.php?from_home=1">
79       <button>Search the Library</button>
80     </a>
81
82
83     <?php if (empty($_SESSION['username'])): ?>
84       <a href="register.php">
85         <button>Create an Account</button>
86       </a>
87       <a href="login.php">
88         <button>Login</button>
89       </a>
90     <?php else: ?>
91       <a href="my_reservations.php">
92         <button>My Reservations</button>
93       </a>
94     <?php endif; ?>
95   </div>
96 </div>
97
98 <div class="feature-grid">
```

```
33 .cta-buttons a {
34     text-decoration: none;
35 }
36
37 .feature-grid {
38     margin-top: 40px;
39     display: grid;
40     grid-template-columns: repeat(auto-fit, minmax(220px,1fr));
41     gap: 20px;
42 }
43
44 .feature {
45     background: #ffffdf;
46     border: 2px solid #d8c9a5;
47     border-radius: 6px;
48     padding: 20px;
49     text-align: center;
50     box-shadow: 0 0 6px rgba(0,0,0,0.08);
51 }
52
53 .feature h3 {
54     margin-top: 10px;
55     font-family: "Cinzel", serif;
56 }
57
58 .feature p {
59     font-size: 15px;
60 }
61
62 /* Small icons without external images */
63 .feature-icon {
64     font-size: 34px;
65 }
```

db_config.php footer.php header.php home.php X

C: > xampp > htdocs > WebD > CA1 > home.php

```
1  <?php include 'header.php'; ?>
2
3  <style>
4  /* Home Page Styling Only */
5
6  .hero {
7      text-align: center;
8      padding: 30px 16px;
9      background: linear-gradient(180deg, #ffffaf1, #f5f1e6);
10     border: 2px solid #d8c9a5;
11     border-radius: 6px;
12 }
13
14 .hero h2 {
15     font-family: "Cinzel", serif;
16     font-size: 32px;
17     margin-bottom: 10px;
18 }
19
20 .hero p {
21     font-size: 18px;
22     max-width: 700px;
23     margin: 0 auto 20px auto;
24 }
25
26 .cta-buttons {
27     display: flex;
28     gap: 12px;
29     justify-content: center;
30     flex-wrap: wrap;
31 }
32
33 .cta-buttons a {
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63     <li><a href="register.php">Register</a></li>
64     <li><a href="login.php">Login</a></li>
65     <?php endif; ?>
66 </ul>
67 </nav>
68 </div>
69 </header>
70 <main class="container">
71
```

```

34     }
35 </style>
36 </head>
37 <body>
38 <header class="site-header">
39     <div class="container header-flex">
40         <div class="header-title">
41             <!-- Scroll-like emoji -->
42             <svg class="logo-scroll" viewBox="0 0 64 64" xmlns="http://www.w3.org/2000/svg">
43                 <rect x="12" y="10" width="40" height="44" rx="4" ry="4" fill="#f5f1e6" stroke="#8c7460" stroke-width="2"/>
44                 <line x1="18" y1="20" x2="46" y2="20" stroke="#b29762" stroke-width="2"/>
45                 <line x1="18" y1="28" x2="46" y2="28" stroke="#b29762" stroke-width="2"/>
46                 <line x1="18" y1="36" x2="46" y2="36" stroke="#b29762" stroke-width="2"/>
47                 <line x1="18" y1="44" x2="36" y2="44" stroke="#b29762" stroke-width="2"/>
48             </svg>
49             <h1><a href="home.php">Library of Alexandria</a></h1>
50         </div>
51         <nav>
52             <ul>
53                 <li><a href="home.php">Home</a></li>
54
55                 <!-- Search Link -->
56                 <li><a href="index.php?from_home=1">Search</a></li>
57
58                 <?php if (!empty($_SESSION['username'])): ?>
59                     <li><a href="my_reservations.php">My Reservations</a></li>
60                     <li>Logged in as <strong><?php echo htmlspecialchars($_SESSION['username']); ?></strong>
61                     <li><a href="logout.php">Logout</a></li>
62                 <?php else: ?>

```

```
db_config.php  footer.php  header.php X
C: > xampp > htdocs > WebD > CA1 > header.php
1  <?php
2  if (session_status() === PHP_SESSION_NONE) {
3      session_start();
4  }
5  ?>
6  <!doctype html>
7  <html lang="en">
8  <head>
9      <meta charset="utf-8">
10     <title>Library Reservation System</title>
11     <meta name="viewport" content="width=device-width, initial-scale=1">
12     <!-- Google Fonts for ancient vibe -->
13     <link href="https://fonts.googleapis.com/css2?family=Cinzel:wght@400;600&family=Crimson+Text:wght@400;600" rel="stylesheet">
14     <link rel="stylesheet" href="style.css">
15     <style>
16         body { font-family: "Crimson Text", "Georgia", serif; }
17         .site-header h1 { font-family: "Cinzel", "Georgia", serif; }
18         .logo-scroll {
19             width: 32px;
20             height: 32px;
21             margin-right: 8px;
22         }
23         .header-flex {
24             display: flex;
25             justify-content: space-between;
26             align-items: center;
27             gap: 16px;
28             flex-wrap: wrap;
29         }
30         .header-title {
31             display: flex;
32             align-items: center;
33             gap: 8px;
34         }
35     </style>
36 </head>
37 <body>
```

```
db_config.php  footer.php X
C: > xampp > htdocs > WebD > CA1 > footer.php
1  <?php
2  ?>
3  </main>
4  <footer class="site-footer">
5      <div class="container">
6          <p>&copy; <?php echo date('Y'); ?> Library of Alexandria</p>
7      </div>
8  </footer>
9  </body>
10 </html>
11
```

```
db_config.php X
C: > xampp >htdocs > WebD > CA1 > db_config.php
1  <?php
2  define('DB_HOST', 'localhost');
3  define('DB_USER', 'root');
4  define('DB_PASS', '');
5  define('DB_NAME', 'librarydb');
6
7  function db_connect() {
8      $mysqli = new mysqli(DB_HOST, DB_USER, DB_PASS, DB_NAME);
9      if ($mysqli->connect_errno) {
10         die("Database connection failed: " . $mysqli->connect_error);
11     }
12     $mysqli->set_charset("utf8mb4");
13     return $mysqli;
14 }
15 ?>
16
```